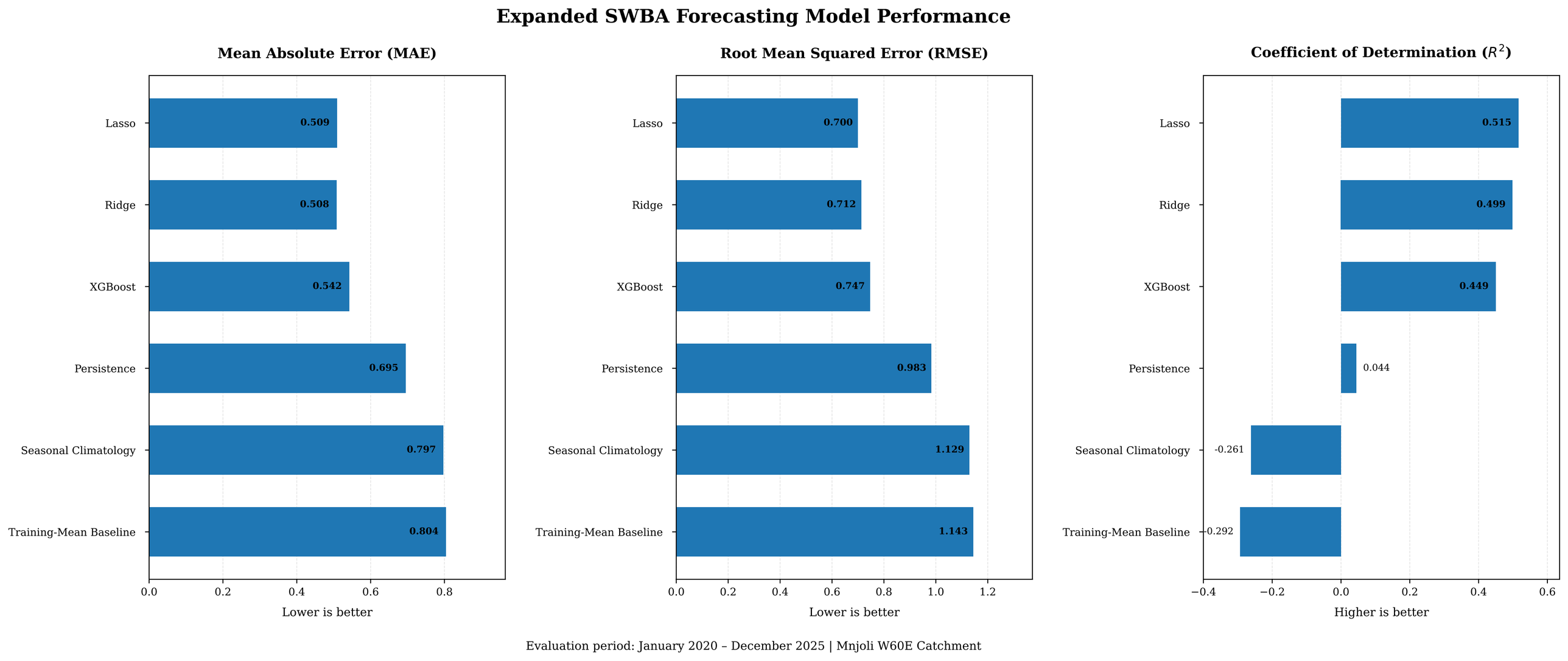
Expanded SWBA Forecasting Model Performance
Mean Absolute Error (MAE)
Lasso
Ridge
XGBoost
Persistence
Seasonal Climatology
Training-Mean Baseline
0.509
0.508
0.542
0.695
0.797
0.804
0.0
0.2
0.4
0.6
0.8
Lower is better
Root Mean Squared Error (RMSE)
Lasso
Ridge
XGBoost
Persistence
Seasonal Climatology
Training-Mean Baseline
0.700
0.712
0.747
0.983
1.129
1.143
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Lower is better
Coefficient of Determination (R^2)
Lasso
Ridge
XGBoost
Persistence
Seasonal Climatology
Training-Mean Baseline
0.515
0.499
0.449
0.044
-0.261
-0.292
−0.4
−0.2
0.0
0.2
0.4
0.6
Higher is better
Evaluation period: January 2020 – December 2025 | Mnjoli W60E Catchment